Doubt-Class

12/09/26 9:06 AM

Bhaiya aap pura padh lo mai saari problems bata deta hoo

First:- kisi bhi variable ko kaha initialise karna hai, upar yaa niche yaa loop me yaa fir loop me kaha

Second:- bade patterns dekhkar samajh nahi aata ki isko kis flow me solve karu

Third:- number pattern me dikkat aati hai that kya karna hai

Fourth:- code ko dry run karna nahi aata so, aap I,j,k aise variables bhi kaise work karte hai bata do

Mera mind stuck ho jaata hai

```
Ques 25:
n = 5

            1
         2  3  4
      5  6  7  8  9
  10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
```

Row=1 Star=1 Space=n-1

while(Row<=n){

//Space

int i=1

while(i<=Space){

}

//Star

int j=1

while(j<=Star){

}

//Row++

Space--

Star+=2

}

Take N as input. For a value of N=7, we wish to draw the following pattern :

```
            1
          2 1   1 2
      3 2 1       1 2 3
  4 3 2 1           1 2 3 4
      3 2 1       1 2 3
          2 1   1 2
            1
```

Row=1, Space1=N-1 Space=-1 Star=1

while(Row<=n){

//Space1

//Star

//Space2

//Star

if(Row<n/2+1){

Space1-=2

Star++

Space+=2

}

else{

Space1+=2

Star--

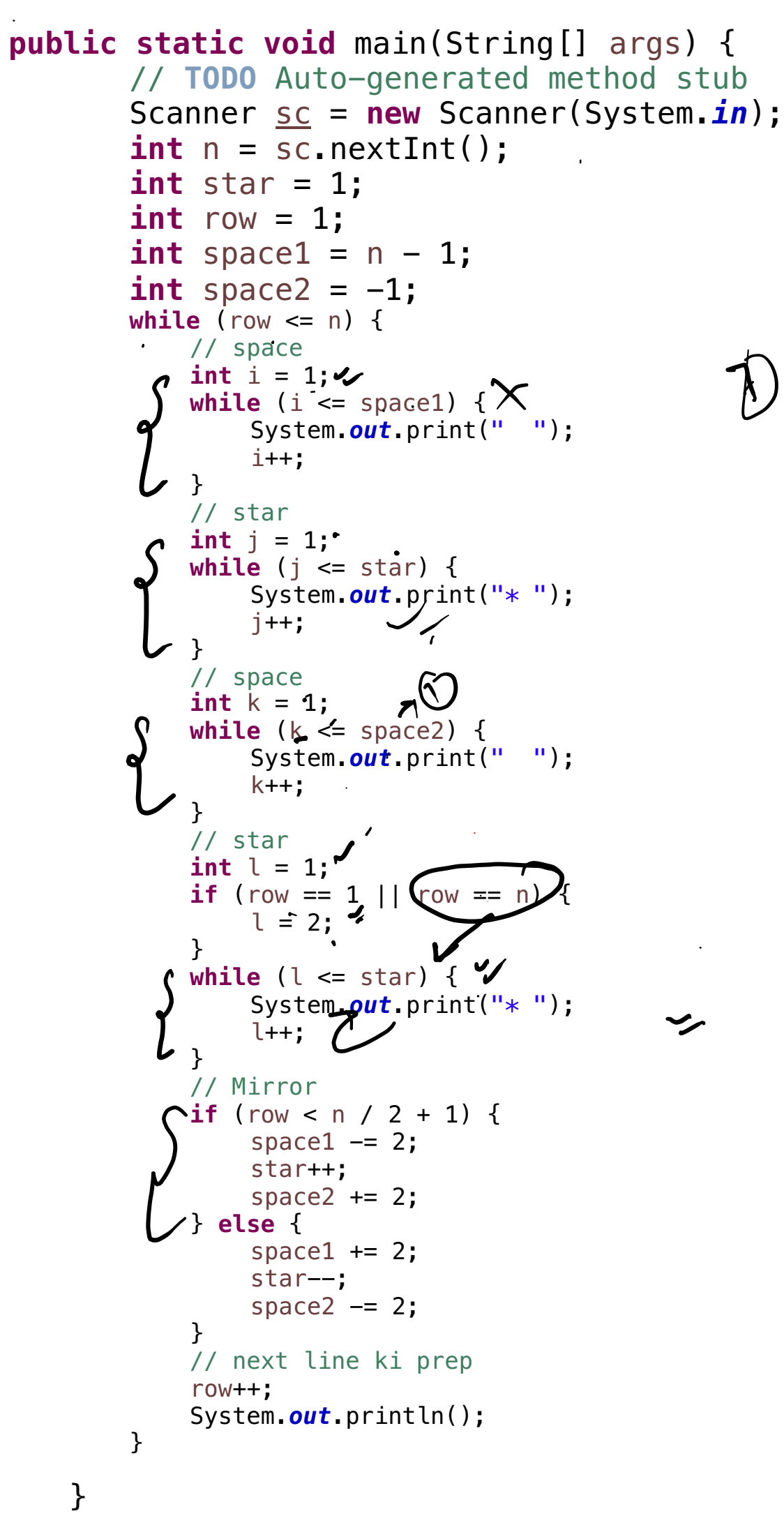
Space-=2

}

Row++

Star

}



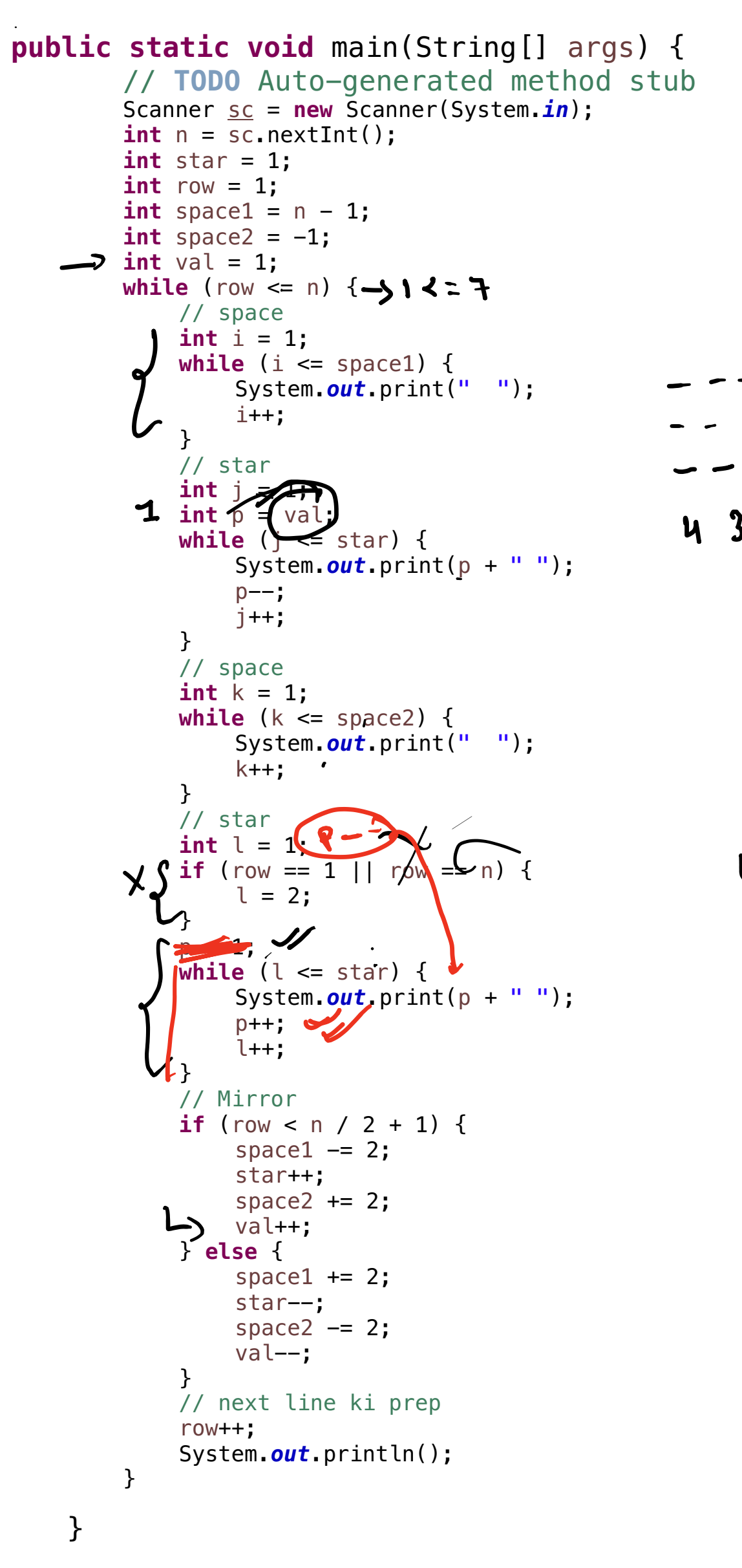
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int star = 1;
        int row = 1;
        int space1 = n - 1;
        int space2 = -1;
        while (row <= n) {
            // space
            int i = 1;
            while (i <= space1) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            while (j <= star) {
                System.out.print("* ");
                j++;
            }
            // space
            int k = 1;
            while (k <= space2) {
                System.out.print("  ");
                k++;
            }
            // star
            int l = 1;
            if (row == 1 || row == n) {
                l = 2;
            }
            while (l <= star) {
                System.out.print("* ");
                l++;
            }
            // Mirror
            if (row < n / 2 + 1) {
                space1 -= 2;
                star++;
                space2 += 2;
            } else {
                space1 += 2;
                star--;
                space2 -= 2;
            }
            // next line ki prep
            row++;
            System.out.println();
        }

    }
```

Row=~~1~~ Space1=~~6~~ ~~4~~ ~~2~~ ~~0~~ Space2=~~-1~~ ~~1~~ ~~3~~ ~~5~~ 3 Star=~~1~~ ~~2~~ 3

7 5 4

```
------*
----*-*
--*-*-**
*-*-*-----*
--*-*-*
----*-*
------*
```



```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int star = 1;
        int row = 1;
        int space1 = n - 1;
        int space2 = -1;
        int val = 1;
        while (row <= n) {
            // space
            int i = 1;
            while (i <= space1) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            int p = val;
            while (j <= star) {
                System.out.print(p + " ");
                p--;
                j++;
            }
            // space
            int k = 1;
            while (k <= space2) {
                System.out.print("  ");
                k++;
            }
            // star
            int l = 1;
            if (row == 1 || row == n) {
                l = 2;
            }
            while (l <= star) {
                System.out.print(p + " ");
                p++;
                l++;
            }
            // Mirror
            if (row < n / 2 + 1) {
                space1 -= 2;
                star++;
                space2 += 2;
                val++;
            } else {
                space1 += 2;
                star--;
                space2 -= 2;
                val--;
            }
            // next line ki prep
            row++;
            System.out.println();
        }

    }
```

1<=7

p--

Row=1 Space1=~~6~~ ~~4~~ 2 0 Space2=~~-1~~ ~~1~~ 3 5 Star=~~1~~ ~~2~~ 3 4 Val=~~1~~ ~~2~~ 3 4

```
------1
----2 1 - 1 2
--3 2 1 --- 1 2 3
4 3 2 1 ----- 1 2 3 4
```

```
      1
    2 3 - 3 2
  3 4 5 - 5 4 3
4 5 6 7 7 6 5 4
```

8

Take N (number of rows), print the following pattern (for N = 6)

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
```

0!=1

1!=1

0 ← 1 → $^0C_0$

1 ← $1 \; ^1C_0$ $\quad 1 \; ^1C_1$

2 ← $1 \; ^2C_0$ $\quad 2 \; ^2C_1$ $\quad 1 \; ^2C_2$

3 ← $1 \; ^3C_0$ $\quad 3 \; ^3C_1$ $\quad 3 \; ^3C_2$ $\quad 1 \; ^3C_3$

4 ← $1 \; ^4C_0$ $\quad 4 \; ^4C_1$ $\quad 6 \; ^4C_2$ $\quad 4 \; ^4C_3$ $\quad 1 \; ^4C_4$

5 ← $1 \; ^5C_0$ $\quad 5 \; ^5C_1$ $\quad 10 \; ^5C_2$ $\quad 10 \; ^5C_3$ $\quad 5 \; ^5C_4$ $\quad 1 \; ^5C_5$

$^{row}C_0$

$val = \frac{row!}{(row-i)!\,i!}$

$^nC_r = \frac{N!}{(N-R)!\,R!}$

$^5C_2 = \frac{5!}{3!\times 2!} = \frac{5\times4\times\cancel{3}\times\cancel{2}\times\cancel{1}}{\cancel{3}\times2\times\cancel{1}} = 10$

AB, AC, AD, AE, BC, BD, BE, CD, CE, DE

A B C D E

#

```
          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1
1   5   10  10  5   1
```

nCR=1

row, C, nCR, nCR+1

$$^NC_R = \frac{N!}{(N-R)!\,R!} \quad \text{—①}$$

$$^NC_{R+1} = \frac{N!}{(N-(R+1))!\,(R+1)!} \quad \text{—②}$$

$$\text{②}\div\text{①} \quad \frac{^NC_{R+1}}{^NC_R} = \frac{\cancel{N!}}{(N-R-1)!\,(R+1)!} \times \frac{(N-R)!\times R!}{\cancel{N!}}$$

$$\frac{^NC_{R+1}}{^NC_R} = \frac{(N-R)!\times R!}{(N-R-1)!\,(R+1)!}$$

$$\frac{^NC_{R+1}}{^NC_R} = \frac{(N-R)\times\cancel{(N-R-1)!}\times R!}{\cancel{(N-R-1)!}\,(R+1)!}$$

$$\frac{^NC_{R+1}}{^NC_R} = \frac{\cancel{R!}\times N-R}{(R+1)\times\cancel{R!}}$$

$\rightarrow (R+1)! = (R+1)(R+1-1)!$
$= (R+1)(R)!$

$\frac{P}{Q} \div \frac{R}{S} = \frac{P}{Q}\times\frac{S}{R}$

$N! = N\times(N-1)!$

$(N-R)! = (N-R)\times(N-R-1)!$

$$\frac{^NC_{R+1}}{^NC_R} = \frac{N-R}{R+1}$$

$$\boxed{^NC_{R+1} = \left[\frac{N-R}{R+1}\right] {^NC_R}}$$

$$\boxed{nCR = \left(\frac{row-i}{i+1}\right) nCR}$$

Take N as input. For a value of N=5, we wish to draw the following pattern :

```
5                   5
5 4               4 5
5 4 3           3 4 5
5 4 3 2       2 3 4 5
5 4 3 2 1   1 2 3 4 5
5 4 3 2 1 0 1 2 3 4 5
5 4 3 2 1   1 2 3 4 5
5 4 3 2       2 3 4 5
5 4 3           3 4 5
5 4               4 5
5                   5
```

Row=1 Star=1 Space=2xN-1

while(Row<=2xN-1){

//Star

//Space

//Star

// mirror

if(Row<n+1){

Star++

Space-=2

}

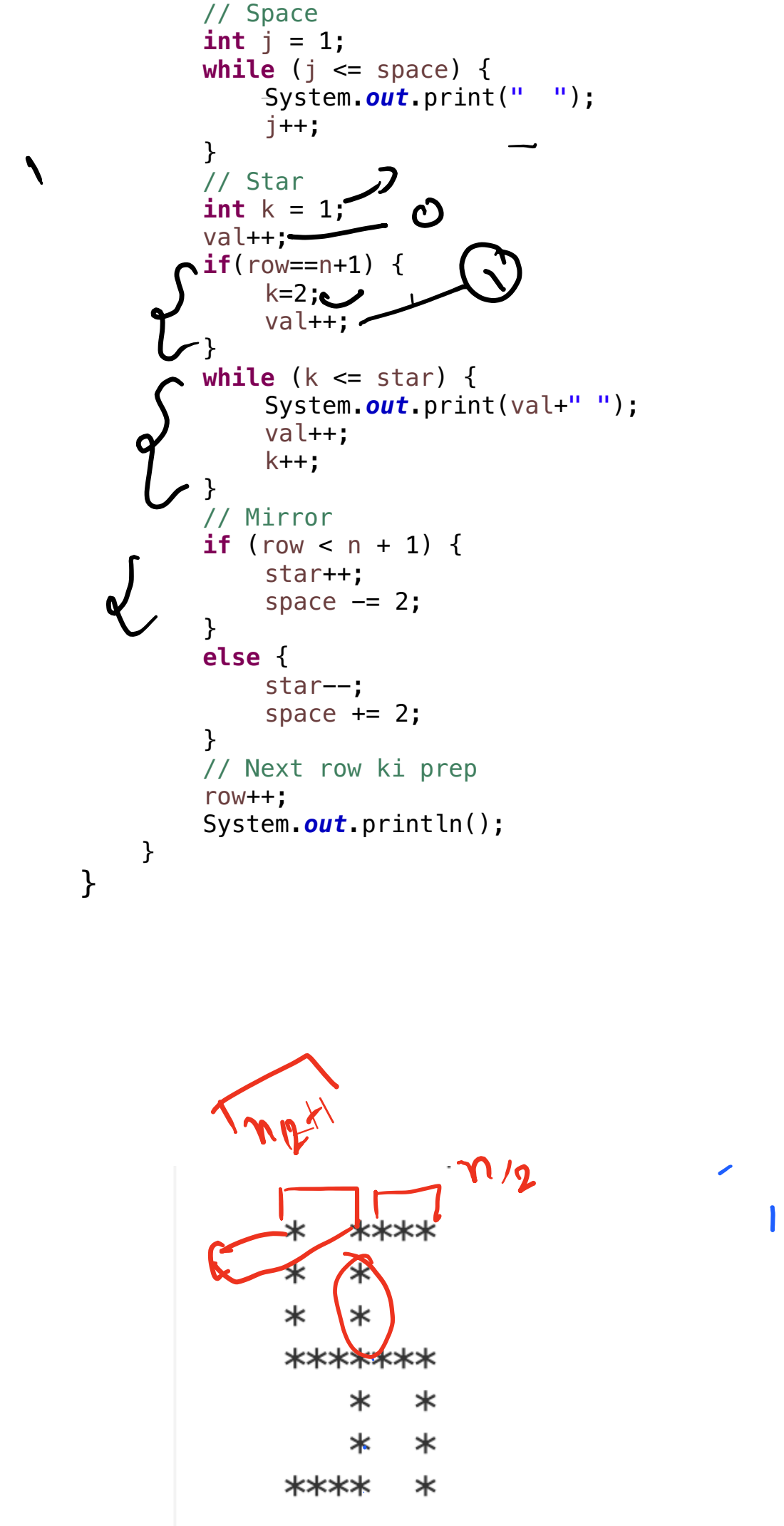
else{ Star--

Space+=2

}

Row++ S+so()

}



```
public class Pattern_InvertedHourGlass {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int star = 1;
        int row = 1;
        int space = 2 * n - 1;
        while (row <= 2 * n + 1) {
            // Star
            int i = 1;
            int val=n;
            while (i <= star) {
                System.out.print(val+" ");
                val--;
                i++;
            }
            // Space
            int j = 1;
            while (j <= space) {
                System.out.print("  ");
                j++;
            }
            // Star
            int k = 1;
            val++;
            if(row==n+1) {
                k=2;
                val++;
            }
            while (k <= star) {
                System.out.print(val+" ");
                val++;
                k++;
            }
            // Mirror
            if (row < n + 1) {
                star++;
                space -= 2;
            }
            else {
                star--;
                space += 2;
            }
            // Next row ki prep
            row++;
            System.out.println();
        }
    }
```

① ② ③ 0 -1

Row=1 Star=~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6 Space=~~9~~ ~~7~~ ~~5~~ ~~3~~ ~~1~~ -1

6 5 4 3 2

```
5 --------- 5
5 4 ------- 4 5
5 4 3 ----- 3 4 5
5 4 3 2 --- 2 3 4 5
5 4 3 2 1 - 5 4 3 2 1
5 4 3 2 1 0 1 2 3 4 5
```

n/2+1

n/2

```
*  ****
*  *
*  *
*******
   *  *
   *  *
****  *
```

```
1
- x x x
x
x x x x x
x x
x x x x
```

n/2+1

```
*  ****
*  *
*  *
*******
   *  *
   *  *
****  *
```

4

n/2+1

n/2

n/2+1

Row

i=1

ci

→n